

TECHNICAL BULLETIN

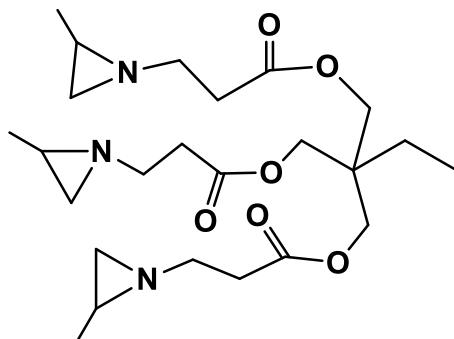
DIXIE CHEMICAL
10601 Bay Area Blvd.
Pasadena, TX 77507
Tel: (281) 474-3271

E-Mail: info@dixiechemicalgroup.com

DCA 467
(Trimethylolpropane tris(2-methyl-1-aziridinepropionate))
CAS Registry Number 64265-57-2

Chemical Properties

DCA 467 contains three aziridine functional groups with an average molecular weight of ~467 g/mole and a functional equivalent weight of ~157 g/mole. It can be used to crosslink polymers with reactive carboxyl functionality at ambient temperatures.



Typical Properties

Physical Appearance	Liquid
Color	Light Yellow
Odor	Amine like
Relative density (water =1)	1.08
Flash Point (ASTM D3828)	487°F (253°C)
Dynamic Viscosity (R.T.)	~ 200 cPs

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Specifications

Property	Values	Test Method
Appearance	Light yellow liquid	D-104 (Visual)
Viscosity at 25°C, cPs	150 - 350	D-150F
Propyleneimine, ppm	< 1.0	D-160Y
Solids, %	99 Min	D-160Y
Color, Gardner	2 Max	D-150G

Applications

DCA 467 is a polyaziridine crosslinker. Addition of one to three percent DCA 467 to acrylic emulsions or waterborne urethane polymers produces a marked improvement in water, chemical, abrasion, and humidity resistance. It is ideal for use with fast air-dry NeoCryl acrylic emulsions, NeoRez waterborne urethane and NeoPac urethane acrylic copolymer products. In addition, DCA 467 can be used with solvent based systems provided reactive carboxyl functionality is present.

The equivalent weight of DCA 467 is 157 g/mole. For many applications, 0.6 equivalents of DCA 467 are sufficient to achieve the optimum improvement in film properties. However, some systems may require the full stoichiometric equivalent of crosslinker.

Key Benefits

- Crosslinked Properties Obtained at Room Temperature
- Improved Water, Alcohol and Print Resistance
- Increased Chemical, Stain and Abrasion Resistance

Toxicity Data

An acute oral toxicity study with DCA 467 produced an LD50 value of 3,038 mg/kg. This places it in a category of toxic materials with a value

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which is similar to common solvents such as methyl ethyl ketone and toluene.

A primary skin irritation value of 2.65 was obtained out of a possible score of 8. Compounds producing a primary irritation value of 5 or greater are defined as "primary skin irritants".

DCA 467 was found to be a primary eye irritant and corrosive effects were indicated.

DCA 467 was tested for potential mutagenic activity in microbial assays using the reverse mutation technique described by Ames. A dose-related increased reversion frequency over the spontaneous rate was obtained and a value of 0.18 revertant per microgram was calculated. Although positive, the level of response is measurable low.

Mixtures of NeoCryl A-601 latex with the recommended amount of DCA 467 were tested immediately upon mixing, and at various elapsed times. None of the mixtures showed any genetic response in the Ames Test. This indicates that the coating vehicle containing DCA 467 before and during applications does not have any measurable mutagenic activity, although the title compound, DCA 467, is mildly active.

Films of NeoRez R-960 and NeoCryl A-601 containing recommended amounts of DCA 467 did not demonstrate any mutagenic activity in any of the assays tested. It can, therefore, be concluded that dried films incorporating DCA 467 as a crosslinker can be considered non-mutagenic by the test conditions described by Ames.

Handling

Care must be taken to avoid any contact with the skin and eyes. When used in spray applications, particular care must be taken to avoid oral and/or nasal ingestion. Eye protection, gloves and suitable clothing must be worn to avoid contact with skin and eyes. In addition, a suitable respirator must be worn for spray applications.

Where spillage of DCA 467 occurs, the area should be thoroughly flushed with water. This will hydrolyze the DCA 467 and make it easier to clean. If more rapid cleaning is desirable, a 5% solution acetic acid or a mineral acid can be used.

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Containers of DCA 467 must show the following warning:

HANDLE WITH CARE

USE ONLY WITH ADEQUATE VENTILATION. AVOID CONTACT WITH SKIN OR EYES AND BREATHING OF SPRAY MIST. IN CASE OF SKIN OR EYE CONTACT, FLUSH THOROUGHLY WITH PLENTY OF WATER. FOR EYES, ALSO CALL PHYSICIAN. CLOSE CONTAINER AFTER EACH USE. KEEP OUT OF REACH OF CHILDREN. STORE IN COOL, DARK PLACE.

For more detailed information on handling and precautions, see the Safety Data Sheet.

Storage Conditions

Store in a cool, dry, dark place. If stored under conditions of excessive heat for extended periods, the material may discolor, deteriorate, and even gel.

Availability

DCA 467 is available from Dixie Chemical in Pasadena, TX. Contact your Dixie representative for details.